

2	(a) (i)	force is rate of change of momentum	B1	[1]
	(ii)	work done is the product of the force and the distance <u>moved</u> in the direction of the force	B1	[1]
	(b) (i)	$W = Fs$ or $W = mas$ or $W = m(v^2 - u^2) / 2$ or $W = \text{force} \times \text{distance } s$	A1	[1]
	(ii)	$as = (v^2 - u^2) / 2$ any subject	M1	
		$W = mas$ hence $W = m(v^2 - u^2) / 2$	M1	
		RHS represents terms of energy or with $u = 0$ $KE = \frac{1}{2}mv^2$	A1	[3]
	(c) (i)	work done = $\frac{1}{2} \times 1500 \times [(30)^2 - (15)^2]$ (=506250)	C1	
		distance = $WD / F = 506250 / 3800 = 133 \text{ m}$	A1	[2]
		or $F = ma$ $a = 2.533 \text{ (ms}^{-2}\text{)}$	C1	
		$v^2 = u^2 + 2as$ $s = 133 \text{ m}$	A1	
	(ii)	the change in kinetic energy is greater or the work done by the force has to be greater, hence distance is greater (for same force)	A1	[1]
		allow: same acceleration, same time, so greater average speed and greater distance		

2	(a) (i)	stress = force / (cross sectional) area	B1	[1]
---	---------	---	----	-----